

- Cover the material with 24 gauge perforated aluminum sheet with 30% perforation center to center with self tapping screws.
- Perforated Aluminum Sheet shall be sealed with 20 mm aluminum beading (25 mm x 3 mm) over all vertical channel frame work. GI screws with washer shall be fixed to horizontal channels. Perforated sheet shall be stretched tight over frame work
- Wherever x/y is mentioned, it indicate AHU room/DG room.

4.0 CHILLED/HOT WATER PIPES INSULATIONS (EPD CERTIFIED):

All chilled water, refrigerant, and condensate drain piping shall be insulated in the manner specified herein. Before applying insulation, all pipe shall be brushed and cleaned. All MS pipes shall be provided with a coat of zinc chromate primer. Thermal insulation shall be applied as follows or as specified in drawings or schedule of quantity:

Pipe nominal bore	Thickness of Nitrile Rubber or Polyethylene foam insulation
Chilled Water Pipes	
15mm – 25mm	19mm
32mm – 80mm	25mm
100mm – 400mm	32mm (19 mm + 13 mm or any equivalent thickness in two layers)
Above 400mm	45mm (32 mm + 13 mm)
Condensate Drain Pipes	
25 mm – 50 mm	13 mm

Insulating material in tube form shall be sleeved on the pipes. On piping, slit opened tube from insulating material shall be placed over the pipe and adhesive shall be applied as suggested by the manufacturer. Adhesive must be allowed to tack dry and then press surface firmly together starting from butt end and working towards centre. Wherever flat sheets shall be used it shall be cut out in correct dimension using correct tools. Scissors or Hacksaw-blade shall not be allowed. All longitudinal and transverse joints shall be sealed as per manufacturer recommendations. All longitudinal and transverse joints shall be sealed by providing 6 mm thick, 50 mm wide nitrile rubber tape. The adhesive shall be strictly as recommended by the manufacturer. The insulation shall be continuous over the entire run of piping, fittings and valves. All valves, fittings, joints, strainers etc. in chilled water piping shall be insulated to the same thickness as specified for the main run of piping and application shall be same as above. Valves bonnet, yokes and spindles shall be insulated in such a manner as not to cause damage to insulation when the valve is used or serviced.

Manufacturer's installation manual shall be submitted and followed for full compliance. All insulation work shall be carried out by skilled workmen specially trained in this kind of work. All insulated pipes shall be labeled (S.R. or R.R.) and provided with 300 mm wide band of paint along circumference at every 1200 mm for colour coding. Direction of fluid shall also be marked. Un-insulated MS pipes shall be painted throughout and direction of fluid marked. All painting shall be as per relevant BIS codes.

4.1 PROTECTIVE COATING OVER INSULATION

To provide mechanical strength and protection from damage all pipe / duct insulated with nitrile rubber as indicated in BOQ shall be covered with glass fibre cloath of weight 200 GSM and 7 mil minimum thickness or self adhesive fibre glass cloth of 7 mil. thickness.